

JK Tyre BTP — Passenger-Car Radial Pilot

A deterministic, BOM-driven single-pass scheduler that converts a fixed May curing plan into a fully time-stamped, machine-assigned upstream schedule across the eight in-scope PCR components.

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Team **Data Science Team**

Readiness

■ Concept ● **Prototype** ■

Production

Scope **V1 — Demand Fulfilment**

01 — PROBLEM & SOLUTION

Problem Statement

- **Aging windows on every BOM edge** — calendered ply unsafe to build with after 24–48 h; master compounds behave differently outside their window.
- **7-level deep BOM**, eight mandatory components AND-joined at Tyre Building — spreadsheets cannot enforce this consistently.
- **Curing press 14811 is the fixed bottleneck** — once published, the plan is immovable; unfed blocks cascade across the 14-day horizon.
- **No deterministic audit trail** today — re-runs drift, infeasibilities are silently fixed, planners can't reproduce yesterday's decisions.

Solution

- **Forward, single-pass, BOM-driven scheduler** — walks BOM downward from each curing block, sizes lots with MPQ + aging-aware aggregation, dispatches in topological order with a CPM backward pass.
- **Hard constraints enforced, never silently fixed** — aging-MIN/MAX inclusive (L22), MPQ split, machine eligibility, atomic AND-join across 8 components, FEFO with soft reservation (L16/L19).
- **Flag-and-continue (L11)** — curing is never shifted; infeasibilities + aging breaches logged with binding constraint named.
- **Reproducible by construction** — single integer-minute domain (L20), single ceil rounding; byte-identical re-runs verified by integration tests.

02 — VALUE CHAIN COVERAGE

Mixing	Final Mixing	Calendering (FRC)	Belt + Ply Cutter	Extrusion	Bead + Filling	Tyre Building (6001)	Curing (14811 fixed)
Full	Full	Full	Full	Full	Partial	Full	Fixed

■ Full coverage ■ Partial coverage ■ Out of scope / fixed

03 — QUANTIFIABLE BENEFITS

92.3%

OTIF — Building → Curing

36 / 38 productive blocks on time on the reference pilot run (V1 forward pass).

100%

Audit & reproducibility

Byte-identical re-run on dag.json + DataFrame-equal on the 11-sheet workbook; 215+ tests.

0 silent

Defaults — every irregularity logged

1 HALT + ~12 WARN findings surfaced from real input data; no value silently imputed.

Additional advantages

- **Auto-tune anchor (L17)** with guardrail HALT — refuses to commit lots if the longest BOM aging chain cannot reach the first curing block.
- **Eleven-sheet bundled workbook** — schedule, machine view, building-to-curing, aging violations, infeasibilities, reservation log, KPIs, audit findings — diff-friendly for planner review.
- **Stable HALT exit codes (10, 11, 12, 20, 30)** — CI / wrapper scripts branch on the binding finding without scraping logs.
- **Modular pipeline of 13 routes** — each step runs in isolation given the upstream typed result; debugging is incremental, not all-or-nothing.
- **Capstrip subtree (L12) auto-excluded** — out-of-scope BOM branch tagged in the visualisation; engine never tries to schedule against incomplete data.

04 — WHO BENEFITS & TECH STACK

Primary beneficiaries

- **Plant Planning Team** — owns the May curing plan and the upstream feasibility check.
- **Production Scheduling & Floor Supervisors** — execute the schedule by machine view.
- **Industrial Engineering** — reads aging violations + infeasibilities to spot bottlenecks.
- **Data Science / Analytics** — extends to V2 (changeover, utilisation, event-heap dispatch).

Tech stack / tools

- Python ≥ 3.11 | uv package manager | pytest + pytest-cov
- pandas | networkx | openpyxl
- matplotlib | plotly | graphviz | PyYAML
- **Spec discipline:** CLAUDE.md (23 locked decisions, 26-step flow, 5 golden fixtures).

05 — ASSUMPTIONS, CONSTRAINTS & NEXT STEPS

Assumptions & Constraints

- Curing plan is immovable (L4.5) — engine never reschedules curing.
- Aging Master is authoritative; Buffer Master ignored (L4).
- Zero starting GT inventory; raws assumed bottomless pre-horizon (L2).
- Capstrip chain on ice (L12) — corrected Aging / MPQ data pending.
- **V1 changeover = 0 min** (L8); 15-min different-product model deferred to V2.
- Uniform 95 % efficiency factor across all operations (L10).
- Topological greedy dispatch in V1; strict event-heap (L21) deferred to V2.

Next Steps / Ask

- **Planner sign-off** on the OTIF 92.3 % reference run and the 4 logged aging breaches.
- **Supply `proc\_time` for `BD-12843443-4` Filling** — current HALT (exit 10) is by design (§8.D).
- **Corrected Capstrip Aging Master + MPQ** so L12 can be lifted in V2.
- **V2 scope sign-off:** changeover modelling, LSF event dispatch, utilisation objective, same-product clustering.
- **Peer review** by plant engineering lead before the floor pilot.
- **Pilot the floor view** — extend Gantt from 3 sample blocks to all 42 if planner needs it.